STACKS AS A DATA STRUCTURE

1. Real-World Applications of Stacks

Stacks are used in various real-world applications, including:

- **Undo and Redo Operations:** Used in text editors to reverse or restore previous actions.
- **Web Browser History:** Helps users return to previously visited pages.
- **Function Calls:** Programming languages use stacks to manage function calls and returns.
- **Expression Evaluation:** Used by calculators and compilers to evaluate mathematical expressions.
- **Backtracking:** Used in maze-solving programs and games to return to previous steps.
- **Syntax Checking:** Used by compilers to check matching brackets and parentheses in programming code.

2. Why Stacks Are Necessary

Stacks are necessary because they follow the **Last In, First Out (LIFO)** principle, meaning the last item added is the first item removed. This makes them useful for managing data in reverse order, tracking previous actions, handling function calls, and returning to earlier states efficiently.

Stacks also provide simple operations for adding and removing items, making them suitable for applications that require data to be processed in a particular order.

3. Selected Application: Undo and Redo System

The selected application is an Undo and Redo System developed using the stack data structure. The system allows users to type and edit text, undo recent changes, and redo changes that have been undone. It uses two stacks: an Undo Stack and a Redo Stack.

Whenever a user edits the text, the previous state is saved in the Undo Stack. When the user clicks Undo, the most recently saved state is retrieved. When the user clicks Redo, the system restores the most recently undone state.

4. Advantages of the System

- Allows users to correct mistakes easily.
- Reduces the need to rewrite deleted or modified text.
- Demonstrates the practical application of the LIFO principle.
- Makes text editing more convenient and efficient.
- Helps users understand how stacks operate in real-world applications.

5. Conclusion

Stacks are important data structures used in many computing applications. Their Last In, First Out principle makes them suitable for managing actions and retrieving previous states. The Undo and Redo System demonstrates how stacks can be used to make text editing easier by allowing users

to reverse and restore changes.